

Core Python Interview Q&A;

Q: What is Python?

A: Python is a high-level, interpreted, and object-oriented programming language.

Q: What are Python's main features?

A: Easy to learn, interpreted, object-oriented, and has large community support.

Q: What are Python data types?

A: int, float, str, bool, list, tuple, set, dict.

Q: Difference between list and tuple?

A: List is mutable, Tuple is immutable.

Q: What is PEP 8?

A: PEP 8 is the style guide for writing clean Python code.

Q: What is lambda function?

A: A small anonymous function written in one line using 'lambda' keyword.

Q: What are *args and **kwargs?

A: *args allows variable number of positional args, **kwargs for keyword args.

Q: What is Python memory management?

A: Python uses garbage collection and reference counting.

Q: What are decorators?

A: Functions that add extra functionality to another function.

Q: What is difference between deep and shallow copy?

A: Shallow copies outer object; deep copies inner objects too.

Advanced Python Interview Q&A;

Q: What is GIL?

A: Global Interpreter Lock allows only one thread to execute at a time in Python.

Q: What are generators?

A: Functions using 'yield' that return one value at a time.

Q: What are closures?

A: Functions that capture variables from their enclosing scope.

Q: What is monkey patching?

A: Modifying or extending code at runtime.

Q: What is pickling?

A: Converting Python object into byte stream for storage.

Q: Difference between threading and multiprocessing?

A: Threading shares memory; multiprocessing uses separate memory.

Q: What are Python properties?

A: Methods used like attributes via @property decorator.

Q: Explain context managers.

A: They manage resources like files using 'with' statement.

Q: What is metaclass?

A: A class that defines how other classes behave.

Q: What is iterator protocol?

A: An object must implement `__iter__()` and `__next__()` methods.

Flask Interview Q&A;

Q: What is Flask?

A: Flask is a micro web framework written in Python.

Q: What is WSGI?

A: Web Server Gateway Interface, a standard for Python web apps.

Q: What is Jinja2?

A: Flask's templating engine used for dynamic HTML.

Q: How to create a Flask route?

A: Use `@app.route('/path')` decorator above a function.

Q: How to run Flask app?

A: Use 'flask run' or `app.run()` method.

Q: What is session in Flask?

A: Used to store data across requests for a user.

Q: What is blueprint in Flask?

A: Used to organize app into components for modularity.

Q: Difference between Flask and Django?

A: Flask is lightweight; Django is full-featured framework.

Q: What is Flask extension?

A: External libraries that add functionality like login, database, etc.

Q: How to handle errors in Flask?

A: Use error handlers like `@app.errorhandler(404)`.

Django Interview Q&A;

Q: What is Django?

A: Django is a high-level Python web framework for rapid development.

Q: What are Django's main features?

A: ORM, admin panel, security, scalability, and template engine.

Q: What is MVT architecture?

A: Model-View-Template, Django's design pattern.

Q: What is Django ORM?

A: It allows database operations using Python instead of SQL.

Q: How to create a Django project?

A: Use `'django-admin startproject projectname'`.

Q: How to create a Django app?

A: Use `'python manage.py startapp appname'`.

Q: What is migration in Django?

A: Used to apply changes in models to the database.

Q: What is middleware?

A: Middleware is code that processes requests and responses globally.

Q: What is difference between `get()` and `filter()`?

A: `get()` returns one object; `filter()` returns queryset.

Q: How to handle static files in Django?

A: Define `STATIC_URL` and use `{% static %}` in templates.